Face Recognizer

User Interface Manual

Version 1.0 · April 2026

REAL-TIME	AI-POWERED	HUMAN-IN-THE-LOOP	SELF-IMPROVING

This manual provides a complete guide to the Face Recognizer dashboard — covering every workflow from registering a new person to verifying recognition accuracy and flagging misclassifications for automated retraining.

Contents

1. Overview

Face Recognizer is a real-time face recognition dashboard running in your browser. It uses a deep-learning embedding model to identify registered individuals from a live camera feed. The interface is split into two areas: the full-screen video panel on the left and a control sidebar on the right.

1.1 Interface Layout

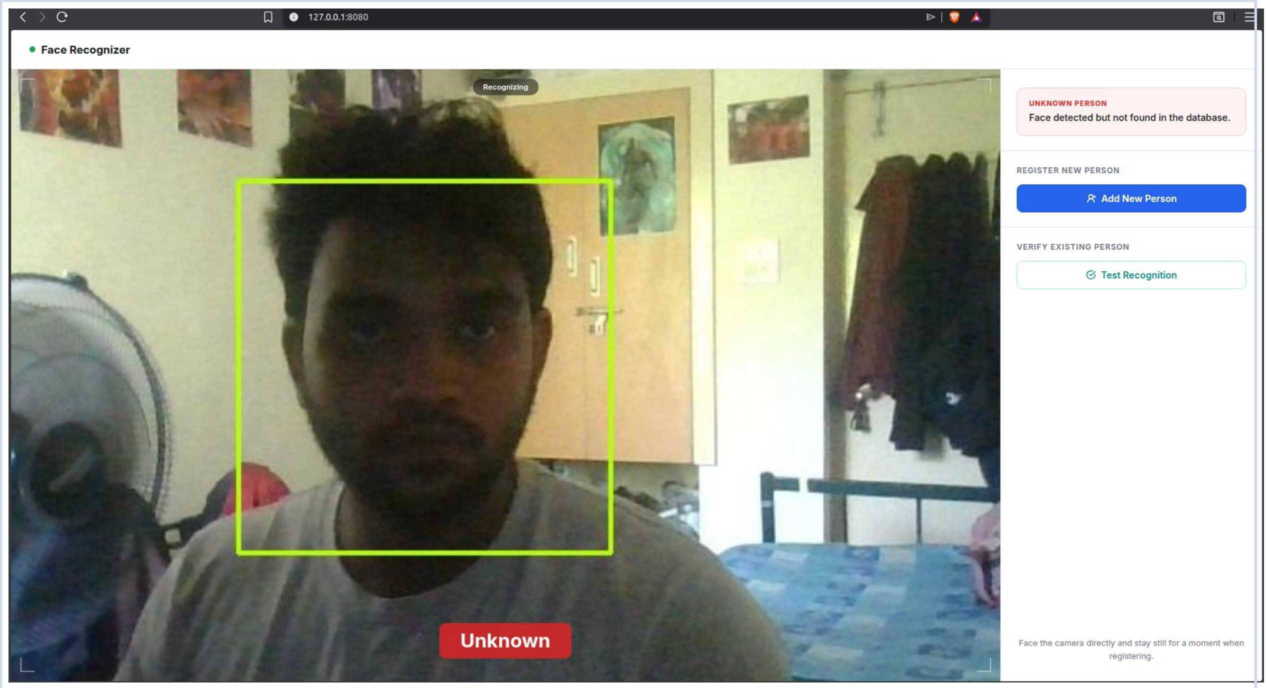

Figure 1 — Main dashboard in Recognizing mode. Live video fills the left panel; controls and status are in the right sidebar. The green bounding box indicates a detected face.

Area	Description
Header bar	App name with a pulsing green live dot.
Video panel	Live camera feed. Green bounding box appears around detected faces. Corner brackets frame the viewport.
Mode badge	Pill-shaped label centred above the video. Changes colour and text to reflect the current mode.
Result overlay	Large coloured name label at the bottom of the video shows the recognised person and confidence score.
Status card	Top of sidebar. Colour-coded: green=ok, amber=warning, red=error, purple=action needed, teal=verify.
Register section	Controls for adding a new person to the face database.
Verify section	Controls for testing recognition on an already-registered person.

1.2 Status Indicators

The status card updates automatically every 600 ms:

GREEN	**Scanning / Recognised** — System is idle or has successfully identified a face.
AMBER	**Registering / Warning** — A face capture is in progress or a non-critical issue occurred.
RED	**Unknown / Error** — Face not found in the database, or a system error occurred.
PURPLE	**Action Required** — Post-registration verification is waiting for your input.
TEAL	**Verify Mode** — Standalone recognition test mode is active.

1.3 Mode Badges

The badge at the top of the video changes to reflect what the system is doing:

Recognizing — Default idle — model is running inference continuously.

Registering: Name — Face capture in progress for the named person.

Verify Recognition — Post-registration HITL verification step is active.

Verify Mode — Standalone verify panel is open.

2. Getting Started

2.1 Launching the Application

Start the full stack and open your browser at:

http://127.0.0.1:8080

The live video feed loads automatically. If the camera cannot be accessed, a dark placeholder frame appears with an error message.

2.2 System Requirements

Component	Requirement
Camera	USB or built-in webcam at /dev/video0
Browser	Chrome, Firefox, or Brave (any modern browser)
Network	Local only — no internet connection required
Model	Trained model in MLflow registry (Production stage)
Docker	All services started via: docker compose up -d

3. Registering a New Person

Registration captures a face embedding and stores it in the database. Once registered, the system can identify that person in real time. The full flow takes about 5–10 seconds.

3.1 Opening the Registration Form

Ensure your face is clearly visible and well-lit, then click the blue **Add New Person** button in the sidebar.

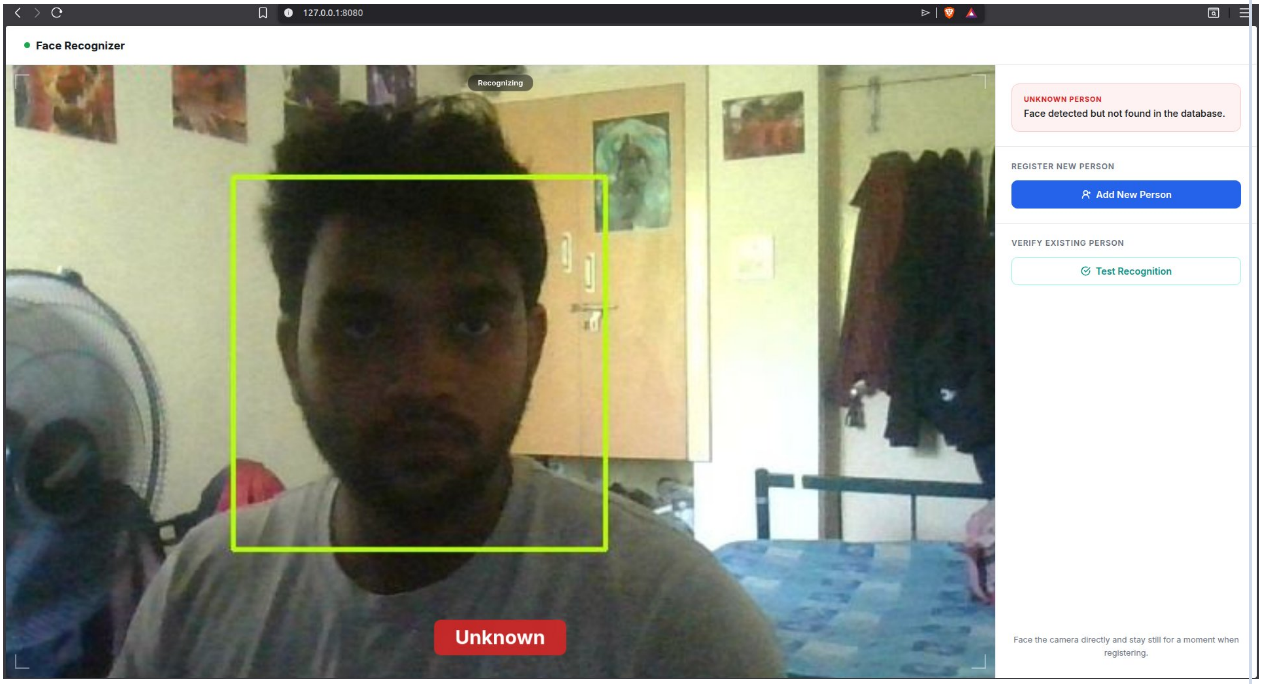

Figure 2 — Dashboard in idle state. 'Add New Person' is visible in the sidebar. The system detects the face but labels it 'Unknown' (not yet registered).

3.2 Entering a Name and Capturing

After clicking Add New Person, a text field appears. Type the person's full name and click **Capture & Save** (or press Enter). The system switches to Registration mode and captures your face after a 1-second hold.

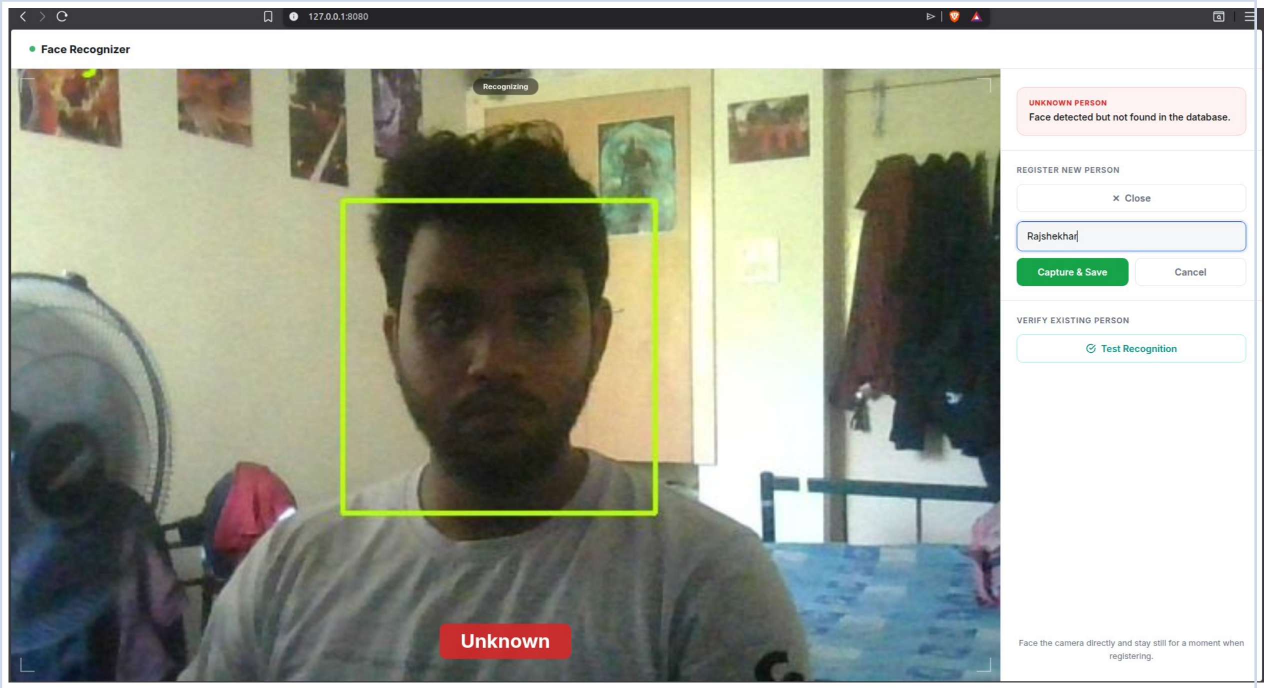

Figure 3 — Registration form expanded with the name 'Rajshekhar' typed. The green bounding box confirms the face is detected and ready to capture.

TIP

Stay still and face the camera directly during the 1-second capture hold. Avoid tilting your head or looking away — this produces a better embedding and improves accuracy.

3.3 Post-Registration Verification (HITL)

After saving, the system enters Human-in-the-Loop (HITL) verification mode automatically. A purple panel appears. Click **Test Recognition** — the model runs one inference and displays the result. If it matches, click **Yes, correct**. If it doesn't, click **No, wrong** to flag it for retraining.

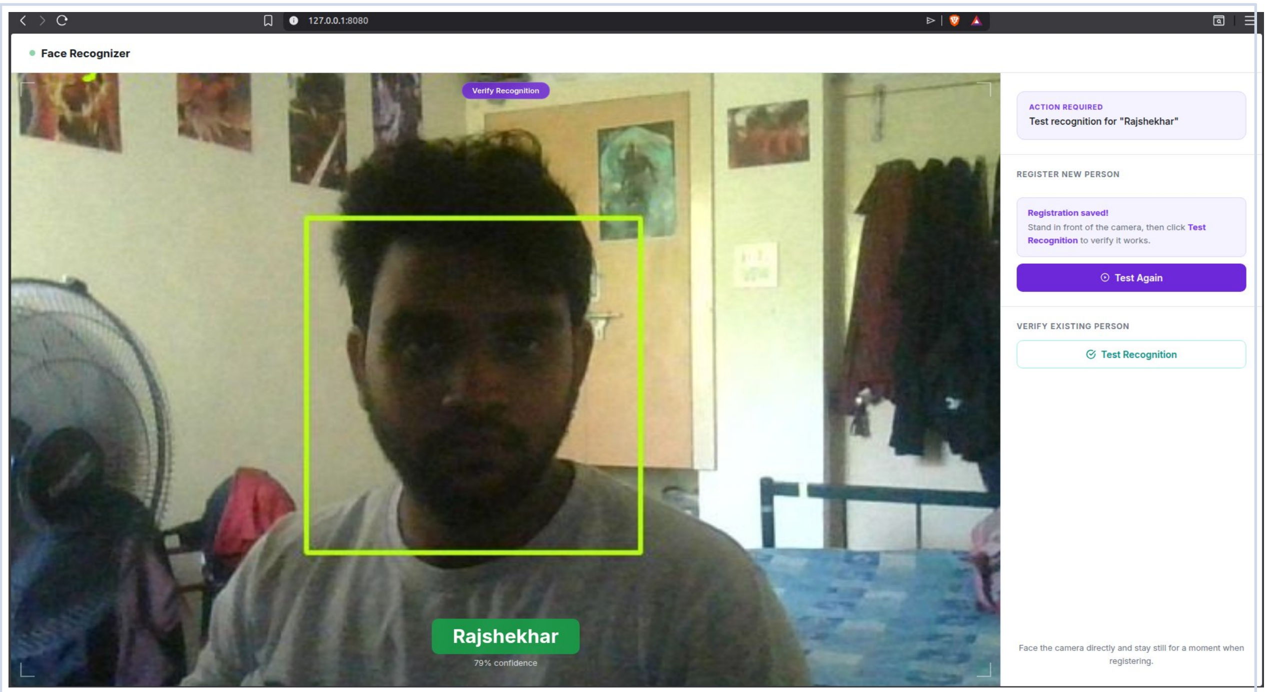

Figure 4 — Post-registration state. Mode badge reads 'Verify Recognition' (purple). The video shows 'Rajshekhar' at 79% confidence after clicking Test Recognition.

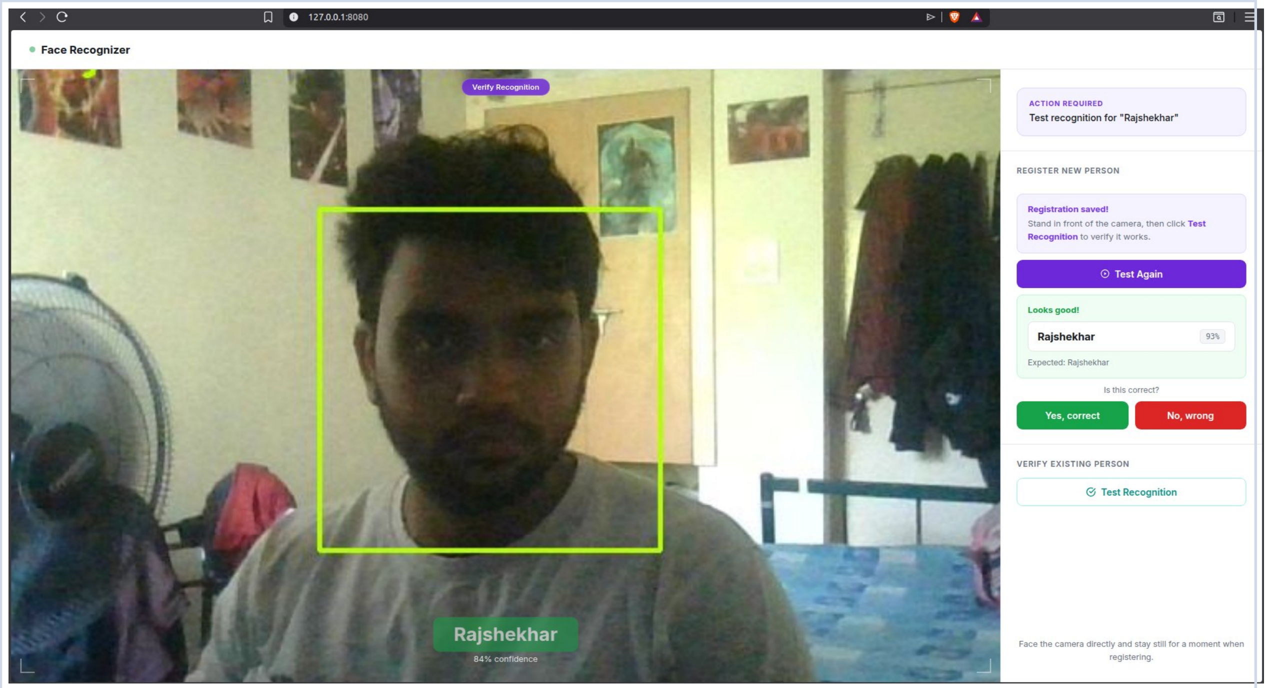

Figure 5 — HITL result panel. Prediction 'Rajshekhar' at 93% matches the expected name. Panel turns green ('Looks good!'). Click 'Yes, correct' to finish, or 'No, wrong' to flag.

Full registration workflow:

1 — **Click 'Add New Person'**
The registration form expands with a name input field.

2 — **Type the full name**
Press Enter or click 'Capture & Save' to begin.

3 — **Hold still (1 second)**
The system captures your face embedding and saves it to the database.

4 — **Click 'Test Recognition'**
Runs a single inference to verify the registration worked.

5 — **Confirm the result**
Click 'Yes, correct' to finish or 'No, wrong' to flag a correction.

4. Verifying an Existing Person

The Verify section lets you test recognition on an already-registered person without going through the full registration flow. Useful after a model update or to spot-check accuracy.

4.1 Opening Verify Mode

Click the teal **Test Recognition** button under "Verify Existing Person". The sidebar expands with a teal instruction panel and a Run Test button. The mode badge changes to 'Verify Mode' (teal).

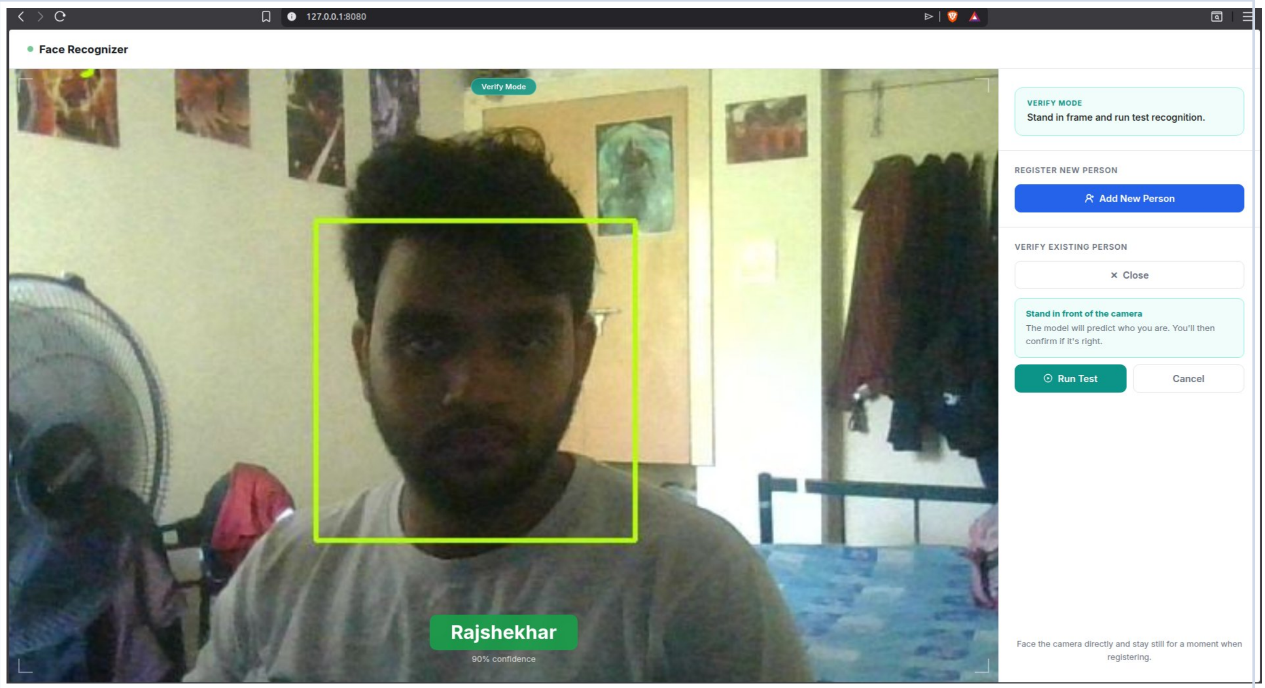

Figure 6 — Verify Mode active. The teal mode badge and 'Stand in front of the camera' panel are visible. The video already shows 'Rajshekhar' at 90% confidence.

4.2 Running the Test

Stand in front of the camera and click **Run Test**. A spinner appears while inference runs. The prediction panel then shows the model's name and confidence.

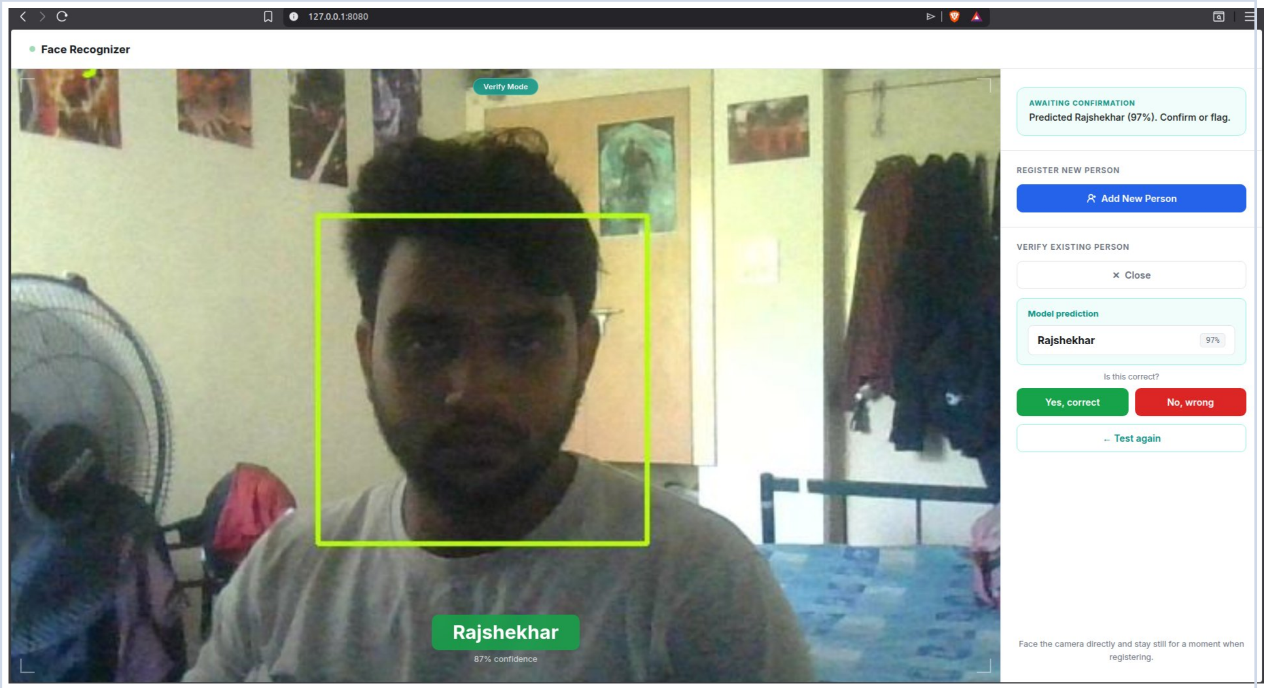

Figure 7 — Verify result. Model predicted 'Rajshekhar' at 97% confidence. The 'Is this correct?' prompt appears with Yes, No, and Test again options.

4.3 Confirming or Flagging

Yes, correct	Closes Verify Mode and shows a green 'Verified' status. No additional data is saved.
No, wrong	Opens a correction form. Enter the actual name and click 'Flag for Retraining'.
← Test again	Returns to Step 1 so you can run another inference without closing Verify Mode.

5. Flagging for Retraining

When the model predicts the wrong person, flag the sample with the correct label. Flagged samples are stored in the database. Once the number of unique pending misclassifications reaches the configured threshold (default: 2), an Airflow DAG is automatically triggered to retrain the model.

5.1 When to Flag

- The model predicts the wrong person's name.
- The model returns 'Unknown' for a registered person.
- Confidence is consistently low (below ~70%) for a known person.

NOTE

Flagging a sample does not immediately retrain the model. Retraining is triggered automatically once enough unique misclassifications accumulate, or it can be triggered manually via the /api/trigger_retrain API endpoint.

5.2 Submitting a Correction

After clicking **No, wrong** in either the HITL or Verify panel:

1 **A red 'Who is this?' panel appears**
You are prompted to enter the correct name for the face.

2 **Type the correct name**
Press Enter or click 'Flag for Retraining' to submit.

3 **Sample is saved**
The face crop and correct label are stored for the next retraining run.

AUTOMATIC RETRAINING

When the misclassification threshold is reached, the system calls the Airflow API to trigger the face_recognition_pipeline DAG. The new model is hot-reloaded into the running application — no restart required.

6. Tips & Best Practices

Good Lighting	Ensure your face is evenly lit from the front. Avoid strong backlight such as sitting with a window behind you.
Face the Camera Directly	Look straight into the lens during registration. Angled captures reduce embedding quality and recognition accuracy.
Stay Still During Capture	Hold still for the full 1-second hold. Motion blur degrades the embedding and leads to lower confidence scores.
Register Multiple Samples	If recognition accuracy is consistently low, register the same person again. Additional embeddings are averaged automatically.
Always Verify After Registration	Complete the HITL verification step after every registration. It confirms the embedding is usable before moving on.
Flag — Don't Ignore Mistakes	Whenever the model is wrong, flag it. Every corrected sample improves the next trained model version.

For technical issues, check container logs with: docker compose logs -f flask-api